

Introduction to Software Engineering

COSC 336

ReSource (Phase I)

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

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Introduction

Concept

ReSource is an intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System designed to support University's asset management department in identifying, sharing, and reuse extra/underutilized assets such as furniture, equipment, vehicles, and books. By doing so a department that requests an asset could use another department's surplus asset and avoid wasteful spending and storage. Furthermore, the system is designed to track each asset's life cycle from its registration, maintenance, transfer, and finally when it's donated/recycled or disposed of. The system combines traditional asset management with AI features such as semantic matching, automated classification, and sustainability driven recommendations.

Document Purpose

This document's purpose is to serve as the phase 1 deliverable which highlights the initial project plan and requirement gathering for ReSource. This document will be the foundation of our project that we will turn to as a reference point to verify that we meet all the requirements stated.

Project Overview

High-Level System Description

ReSource is foundationally built on two layers, a traditional asset management layer and an AI-Powered layer.

Layer 1 Classic Asset Management

Module	Function
Asset Registration	Captures structured asset data such as name, category, condition, quantity, location, estimated value, and images
Resource Marketplace	Publishes registered assets as searchable/filterable listings visible to other departments
Request & Reservation	Lets a department submit a request for an asset specifying purpose, quantity, and urgency
Approval & Transfer	Routes accepted requests to an administrator on approval, updates ownership, custody, and location
Life-Cycle Tracking	Logs maintenance, repair, donation, recycling, and disposal events in the asset's history
Access Management	Assigns role-based permissions, governing which actions each user can perform

Layer 2 AI-Powered Features

Module	Function
AI Matching Engine	Ranks open requests against available assets using semantic similarity plus condition, location, and urgency as inputs
AI Classification	Suggests a category and tags at the point of registration, based on the asset's description
Sustainability Optimizer	Outputs a recommended end-of-life action for the asset whether its reuse, repair, donate, recycle, or dispose
LLM Assistant	Accepts natural language questions and maps them to a search/request action
Generative Reporting	Produces a text summary of a selected time period's activity like transfers, savings, and waste diversions from the dataset

Background and Context

Universities routinely purchase new furniture, computers, laboratory equipment, and office materials, while similar usable items remain unused, stored, or thrown out by other departments. This is often not caused by a shortage of resources, but by the absence of an integrated system to identify, match, and track them.

Nowadays, when a department has surplus assets or needs resources, it's typically handled informally and without a shared system. A department that no longer needs an asset stores it in a stockroom, reports it to facilities for disposal, or occasionally informs other departments by email or word of mouth, without any central listing of what is available. When a department needs a certain asset, it will make a request without checking if another department already has something suitable in surplus. Approvals for transferring assets, when they happen at all, are done through emails or paper forms, with no standardized workflow. Maintenance and repair records, when kept, are usually stored in each department's own spreadsheets or paper logs, making it difficult to see an asset's full history. Decisions about whether to donate, recycle, or dispose of an asset are made case by case, with no consistent criteria and no tracking of the financial and environmental impact.

This manual process of managing assets is unreliable and wastes a significant amount of time and money that could be saved through better coordination. ReSource is designed to replace this outdated, wasteful process with a single, structured platform.

Business Goals and Objectives

- Reduce unnecessary purchases by connecting departments with existing surplus assets.
- Extend the life of campus assets through better tracking and maintenance.
- Increase reuse, donation, and recycling rates across the university.
- Provide insights into the environmental and financial impact of the circular resource management.
- Demonstrate how classic and AI-enhanced features can be combined in a single system.

Expected Benefits

- Financial: Reduced spending by reusing surplus assets and lowering storage costs.
- Operational: clearer accountability through approval workflows and easy to review existing assets.
- Environmental: Reduced waste through donations, recycling, and reusing of assets.
- Organizational: A single source for asset location, condition, and history replacing scattered and informal communication.

Assumptions

- The university would be willing to adopt a centralized platform for internal asset exchange if one were available.
- Departments are willing to accurately register and update the status of their assets.
- Users will have basic computer literacy sufficient to use a web-based system without extensive training.

Project Constraints

- Timeline: The project must be completed within the Fall 2026 semester, following the official Phase 1-8 schedule (September to December 2026).
- Team size: The project is developed by a team of 4 students, with work distributed across planning, requirements, design, and development tasks.
- Scope: As an academic prototype, the system will demonstrate core functionality rather than a production-ready, university-wide deployment.
- Technical resources: AI features (matching, classification) will rely on existing libraries, pretrained models, or APIs appropriate for a semester-length project, rather than custom trained machine learning models.
- Data availability: Real University asset data is not accessible for this course project, so sample or simulated data will be used for development and testing.

Scope

In scope	Out of scope
Core classic modules: asset registration, resource marketplace, request/reservation, approval and transfer workflows, life-cycle history, maintenance/repair logging, and role-based user management	Integration with real university procurement or financial systems
Core AI-enhanced modules: AI-assisted classification, AI-based semantic matching with a basic explanation of recommendations, and a sustainability recommendation feature	Full production-level security certification or compliance auditing
Basic operational and AI-generated reporting	Large-scale, continuously trained AI models; AI components will use existing tools/APIs rather than custom model training

note: a basic natural-language query feature may be included as time allows, but is not a core deliverable

Stakeholders

As stated above, the system will support campus resource reuse, asset tracking, and sustainable decisions. This project involves many users with different authorizations and access limits. Therefore understanding their responsibilities and expectations help in defining suitable features, information access, and authorization control.

The table below identifies all the possible stakeholders, starting from the system developing team to all the different end-users. The table clearly states their roles/responsibilities and their needs/expectations to help build up the system.



Stakeholder (Category)	Role/Responsibility	Needs/Expectations & Data Needs	Feature Requirement Mapping	Permissions & Approval Decisions
Administrators (User)	Manage operational workflows, resolve allocation conflicts, approve transfers or end-of-life actions	Approval queues, operational reports, records of rules and changes; needs visibility into all pending requests and transfer records	Final Approval and Transfer Management with User and Access Management	Approve/reject asset transfers, end-of-life actions, and manage user roles
Procurement Team (User)	Check stock before purchasing and accept or reject requests	Inventory and demand visibility, compatibility information, purchase records; needs inventory levels and purchase-request history	AI Matching Engine giving AI-Generated Reports	View inventory and requests and approve whether a new purchase is justified or if we can reuse surplus assets
Department Representative (User)	Identify department needs and surplus assets with coordinated requests and transfers	Asset inventory and request status since they need visibility into their own department's inventory and requests	Request & Reserve New Asset	Create/edit own department's listings and submit requests while approval sits with Administrators
Asset Coordinators (User)	Register assets and maintain condition, ownership, and location records plus verify availability and review suggested matches	Accurate asset histories, current availability, clear specifications, relevant matches, status updates	Asset Registration & Publication, AI Matching, Sustainability Optimization Resource Marketplace, and Life-Cycle Tracking	Register/update assets and confirm receipt/condition on completed transfers
University Management (Customer)	Represent the university, confirm project priorities and budget authority, sign off on major decisions	Accurate asset records, reduced spending, clear high-level reporting with needs summary-level cost and impact data, not operational detail	Generative AI Reporting (financial savings, waste diversion, avoided purchases)	Final authority on budget and adoption without day-to-day operational permissions
Developers (Software Developer)	Gather/understand requirements, design, build, and test the system	Clearly stated requirements and scope, defined interfaces, access to development tools/environment	Not feature-specific because they are responsible for implementing all classic and AI modules	Full read/write access to codebase and dev environment without business-decision approval authority
Lab Instructors (Development Manager)	Provide academic guidance, review phase deliverables, assess progress against course requirements	Regular, clear deliverables and documentation visibility into evaluation criteria and rubric alignment across phases	Not feature-specific since they governs project process and academic evaluation, not system functionality	Approve/grade phase deliverables without permissions within the deployed system itself

These stakeholder needs will guide the system's development and evaluation.

Requirements Gathering

Our initial requirements for ReSource were identified from the slides and project description. Then we related them to the stakeholder responsibilities and expectations above. The aim is to define how the system will support asset reuse, requests, approvals, life cycle tracking, and AI-assisted decisions at the University.

Interviews and surveys are required to validate these requirements. Below is a table of our requirement gathering methods.

Requirements Gathering Methods

Method	Participants and purpose	Expected output
Document review	Developers review the course materials and the stated ReSource scope.	Initial functions, constraints, and evaluation needs.
Interviews	Administrators, procurement, department representatives, and asset coordinators explain current asset and request processes.	Required data, approval responsibilities, request rules, and common problems.
Short survey	Department representatives and asset coordinators describe search difficulties, useful filters, and expectations of AI suggestions.	User priorities and usability expectations.
Workflow and record walkthrough	Developers and authorized users follow sample listings, requests, approvals, transfers, and maintenance records.	Validated workflow steps, exceptions, and data requirements.
Requirements review	Developers review the baseline with relevant stakeholders; lab instructors review the academic deliverable.	Agreed scope, priorities, success targets, and unresolved questions.

Questions to Confirm

The team will confirm which assets may be shared, who can publish and request them, how procurement reviews requests, and which decisions require administrator approval. Other questions concern reservation expiry, competing requests, receipt confirmation, maintenance evidence, and the information needed to estimate savings.

Summary of gathered requirements

ReSource requires a shared inventory that separates assets in use from the extra assets that are not in use. It also requires a search and request process, and controlled approvals and transfers. Asset condition, ownership, location, and maintenance history must be saved in the system to ensure life-cycle tracking.

The AI layer will support asset classification, resource matching, sustainability optimization, natural language interaction, and generated AI reports. The users must understand and review the AI suggestions.

Initial Requirements Specification

The initial requirements for ReSource follow the five classic functions and five AI functions in the Phase 1 slides. They define the main system behavior; detailed fields, workflow rules, and acceptance conditions will be refined during requirements analysis.

Prioritization and Requirements Review

The core inventory, request, access, approval, and transfer functions receive first priority, followed by AI classification, matching, and reporting. The assistant and sustainability recommendations remain in scope and depend on the core records and workflows. In line with Waterfall, the team will review the requirements before system design and document any later changes.

Functional Requirements

Asset Registration and Publication

ReSource shall allow asset coordinators to register and update campus assets and publish surplus or underutilized resources. Records shall include identifying details, category, quantity, department, owner, condition, location, availability, purchase information, estimated value, and supporting images or documents.

Resource Request and Reservation

ReSource shall allow department representatives and authorized asset coordinators to search and filter available assets, submit departmental needs, reserve suitable resources, and track requests. Requests shall describe the required specifications, quantity, condition, urgency, location, and date. Reservations shall not exceed available quantities.

Approval and Transfer Management

ReSource shall support procurement review of requests and administrator approval of transfers and end-of-life actions within their assigned authority. It shall record approval

decisions, collection or delivery arrangements, receipt confirmation, and changes in ownership, custody, and location.

Inspection and Asset Life Cycle Tracking

ReSource shall record each asset's history from registration and use through inspections, maintenance, repairs, transfers, refurbishment, donation, recycling, retirement, or disposal. Asset coordinators shall record service actions, costs, and condition updates, while actions requiring approval remain under authorized control.

User and Access Management

ReSource shall authenticate users and manage accounts with defined roles and permissions. Users shall access records and perform actions only within their assigned responsibilities and departmental permissions. Authorized administrators shall manage accounts and access assignments.

The workflows shall also provide status notifications, audit records, and conventional reports on inventory, requests, transfers, and maintenance. These support the five main functions and the reporting requirements in the project description.

AI Requirements

AI Based Resource Matching

ReSource shall recommend and rank assets for departmental requests using semantic similarity, technical compatibility, condition, quantity, location, and urgency. Matching shall also consider required dates and available cost information, and provide a compatibility score with a clear explanation.

AI Assisted Asset Classification

ReSource shall suggest categories, tags, and standardized descriptions for newly registered assets using available asset information. Asset coordinators shall be able to review and correct suggestions before they update the approved record.

Sustainability Optimization

ReSource shall recommend whether an asset should be reused, transferred, repaired, refurbished, donated, recycled, or retired. It shall consider condition, costs, and available sustainability information, including component reuse where suitable and disposal as a final option. Authorized users shall approve the resulting actions.

LLM Powered Assistant

ReSource shall provide a natural-language assistant for resource searches, request submission, asset-related questions, status tracking, and navigation. The assistant shall follow user permissions and obtain confirmation before submitting or changing a record.

Generative AI Reporting

ReSource shall summarize resource exchanges, maintenance trends, avoided purchases, financial savings, waste diversion, and estimated carbon-emission reductions. Summaries shall use recorded information, agree with conventional reports, and identify assumptions or missing data behind estimates.

AI shall support decisions while the traditional functions enforce permissions, quantities, and approvals. Users shall review AI suggestions, and the team will compare AI matching and summaries with structured search and conventional reports.

Nonfunctional Requirements

Security and Privacy

ReSource shall protect user, asset, and financial information from unauthorized access or changes. Data shared with external AI services shall be limited to information approved for that use.

Usability

ReSource shall provide clear forms, asset availability, and request-status information. At least 80% of test users shall be able to find and request a suitable asset without assistance.

Reliability and Data Accuracy

ReSource shall maintain consistent quantities, ownership, condition, and status records and prevent duplicate reservations. Users shall be able to continue core tasks through ordinary forms and structured search if an AI service fails.

Measurable Project Success Criteria

Measure	Proposed success criterion
Core workflow	All agreed scenarios for registration, publication, requests, approvals, receipt, and maintenance pass. No critical access, quantity, or ownership error remains unresolved.
Availability and tracking	For every asset in the test dataset, displayed availability agrees with its reservation and maintenance status. Every completed test transfer has receipt confirmation and an updated location and custody record.
User acceptance	At least 80% of at least five representative test users find and request a suitable asset without assistance. Report participant roles and actual completion counts.
Response time	With 100 asset records and five simultaneous users, at least 95% of at least 30 structured searches complete within 3 seconds. At least 90% of at least 20 AI interactions return a usable result within 15 seconds; all failures provide a clear fallback.
Match usefulness	For at least 20 reviewed requests with eligible assets, at least 80% have a suitable asset in the top three AI results. Department representatives or asset coordinators judge suitability. Compare with structured search using the same requests.
Classification quality	At least 80% of at least 30 held-out asset examples receive the correct primary category against reviewed labels. Report category coverage and corrections separately.
Reuse and avoided purchases	Demonstrate at least three confirmed transfers that satisfy documented departmental needs. Count a purchase as avoided only when procurement verifies that reuse displaced a planned purchase.
Repair and life cycle tracking	Demonstrate at least two repair cases with defect, repair, cost, condition, and return-to-service records. Each evaluated asset has a complete history of the events performed in the demonstration.
Reports and sustainability	Conventional and AI reports agree with the underlying transfer and cost records. Every reported savings or impact estimate has traceable inputs and assumptions. Report reuse, repair, life extension, diverted mass, and carbon estimates only where the available evidence supports them.

Initial Project Plan Outline

Development Methodology

The ReSource project will follow the Waterfall development methodology. The project will progress through sequential stages. The Waterfall method suits this project because it allows early identification of the main system functions and requirements, so each stage can be completed and reviewed before moving to the next.

Timeline and Milestones

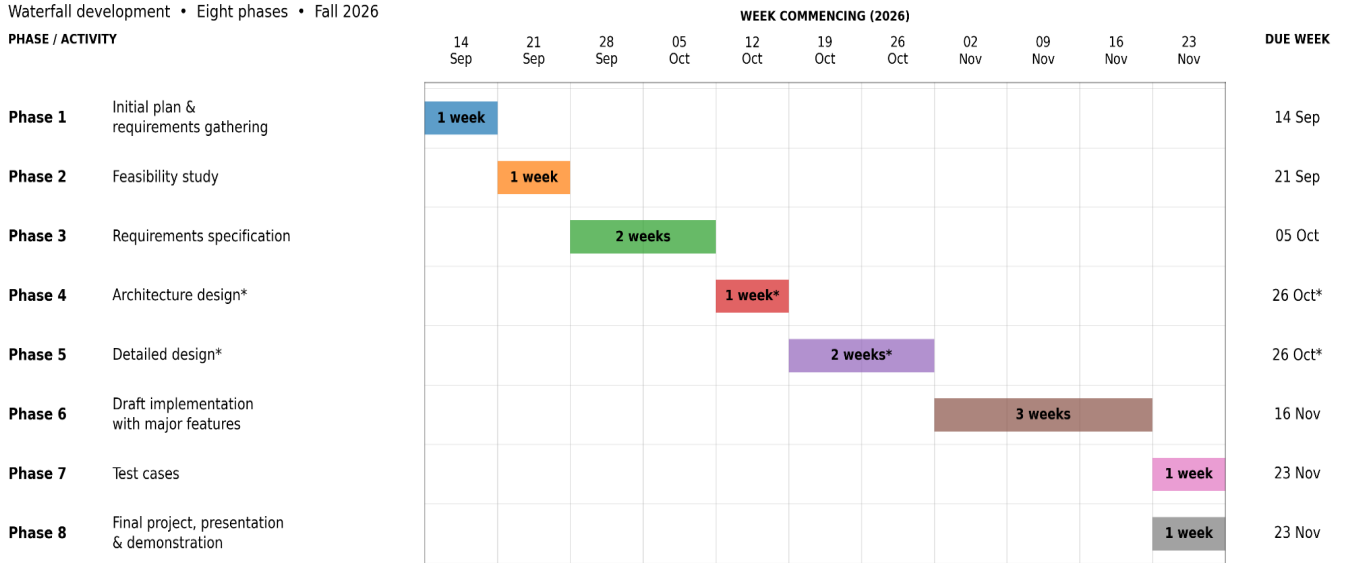
Phase	Main Activities	Milestone	Dependencies
Phase 1 Planning and Requirements Gathering	Define the scope, stakeholders, initial requirements, constraints, and preliminary project plan.	Initial plan and requirement gathering document completed. <i>Due: week of 14 Sep 2026</i>	None. Project initiation.
Phase 2 Feasibility Study	Assess technical, economic, operational, and schedule feasibility, including AI tools and data availability.	Feasibility document completed. <i>Due: week of 21 Sep 2026</i>	Phase 1: Scope and initial requirements.
Phase 3 Requirements Specification	Finalize functional, non-functional, AI, and sustainability requirements, business rules, and acceptance criteria.	Requirements document completed. <i>Due: week of 5 Oct 2026</i>	Phases 1 and 2: Stakeholder needs and feasibility findings.
Phase 4 Architecture Design	Define system layers, modules, interfaces, data flow, and the AI integration approach.	Architecture section completed for the combined design document. <i>Due: week of 26 Oct 2026</i>	Phase 3: Completed requirements specification.

Phase 5 Detailed Design	Specify database schemas, workflows, UML models, user interfaces, and component-level logic.	Combined architecture and detailed design document completed. <i>Due: week of 26 Oct 2026</i>	Phases 3 and 4: Requirements and system architecture.
Phase 6 Implementation	Develop core asset-management functions and planned AI features; integrate major system modules.	Draft implementation with major features completed. <i>Due: week of 16 Nov 2026</i>	Phases 4 and 5: Completed architecture and detailed design.
Phase 7 Testing and Test Cases	Prepare and execute functional, integration, usability, security, and AI test cases; document results and corrections.	Test-case details completed. <i>Due: week of 23 Nov 2026</i>	Phases 3 and 5: Test basis. Phase 6: Prototype for test execution.
Phase 8 Final Documentation and Delivery	Finalize documentation and corrections; prepare the project presentation and system demonstration.	Final project submitted; presentation and demonstration delivered. <i>Due: week of 23 Nov 2026</i>	Phases 6 and 7: Working prototype, test results, and necessary corrections.

The main milestones of the project are the completion of the requirements specification, completion of the system design, development of the working prototype, completion of system testing, and final submission of the system and documentation.

ReSource | Project Gantt Chart

Waterfall development • Eight phases • Fall 2026



* Phases 4-5: proposed 1 + 2 week split. The combined design document is due in the week of 26 October.

Phases 7-8 share the final one-week window and are both due in the week of 23 November.

Source: supplied course schedule. Bars are proposed weekly allocations; exact start dates and lab-day deadlines are not specified.

Risks

The following preliminary risks will be monitored throughout the project. Likelihood and impact are initial team estimates and will be reviewed before Phase 2.

Risk	Likelihood	Impact	Mitigation
Uneven familiarity with AI tools and APIs across team members	Medium	Medium	Pair less-experienced members with experienced teammates on AI tasks. Schedule short training sessions and document setup and integration steps.
Scope creep from adding AI features beyond the agreed requirements	Medium	High	Follow the agreed requirements prioritization and review scope changes before implementation. Treat only agreed optional features as stretch goals.
Irregular GitHub commits leading to poor visibility of progress and individual contributions	Low	High	Commit genuine work incrementally with clear messages and task references. Avoid leaving all changes until one final push.
Inconsistent formatting or tone across sections written by different members	Medium	Low	Use a shared formatting template.

Resource Allocation

The project will require a combination of human, software, hardware, and AI resources.

Resource	Allocation / Purpose
Project Team	Four team members responsible for requirements analysis, design, implementation, testing, and documentation. Fully collaborative, with rotating administrative roles
Development Computers	Used for programming, system design, testing, and demonstrations
Development Tools	IDEs and libraries used to implement the system
Database System	Used to store users, assets, requests, transfers, maintenance records, and asset life-cycle information
Version Control	Git/GitHub or a similar platform for source-code management and collaboration
Modeling Tools	Used for UML diagrams, database models, and system-design documentation
AI Models / APIs	Used for asset classification, semantic matching, recommendations, and natural-language interaction
Test Data	Sample asset, request, maintenance, and sustainability data used for development and evaluation